

Scenario

You are employed in the ICT Department of [Gelos Enterprises](#) (Gelos). Gelos has been contracted by [Indigo Community Services and Health Hub](#) (Indigo) to train and evaluate a machine learning (ML) model to classify health risks of clients.

Work brief

Indigo is a network of health facilities, including health services, a dental clinic and rehabilitation facilities. Indigo is looking to develop a machine learning (ML) model to classify the cancer risk of their clients based on their history and demographics. This classification model will help the healthcare professionals at Indigo to better understand which clients are at high risk and require immediate intervention or closer monitoring.

Your supervisor at Gelos, Morgan Garcia, has asked you to train and evaluate an ML model to automate the task of classifying clients as high, medium or low risk of developing cancer.

Morgan has had good experience with using neural network models in past projects so wants you to use this type. They have suggested using scikit-learn's MLPClassifier model however you may use another neural network model if you prefer (refer to scikit-learn's page [1.17. Neural network models \(supervised\)](#)).

Indigo has provided some client data for you to review and use to build the ML model. Morgan has asked you to prepare the data as follows:

- Categorise ages into the following groups:
 - 0-14
 - 15-24
 - 25-34
 - 35-44
 - 45-54
 - 55-64
 - 65-74
 - 75-84
 - 85 +
- Split the data by using 70% for training, 15% for validation and 15% for testing.

Indigo would like the accuracy rate of the ML model to be 90% or higher.